

Mastering the mainframe

A developer in Wonderland: An Adventure in Coding

Wolfgang Beikircher

Mainframe Development Lead

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Agenda

Intro - What the Hell is a Mainframe?

Programming the Shark: The Basics

The Mainframe Lives! Trends and Innovations

Numbers and Insights: Data and Statistics

Q&A: Questions & Answers



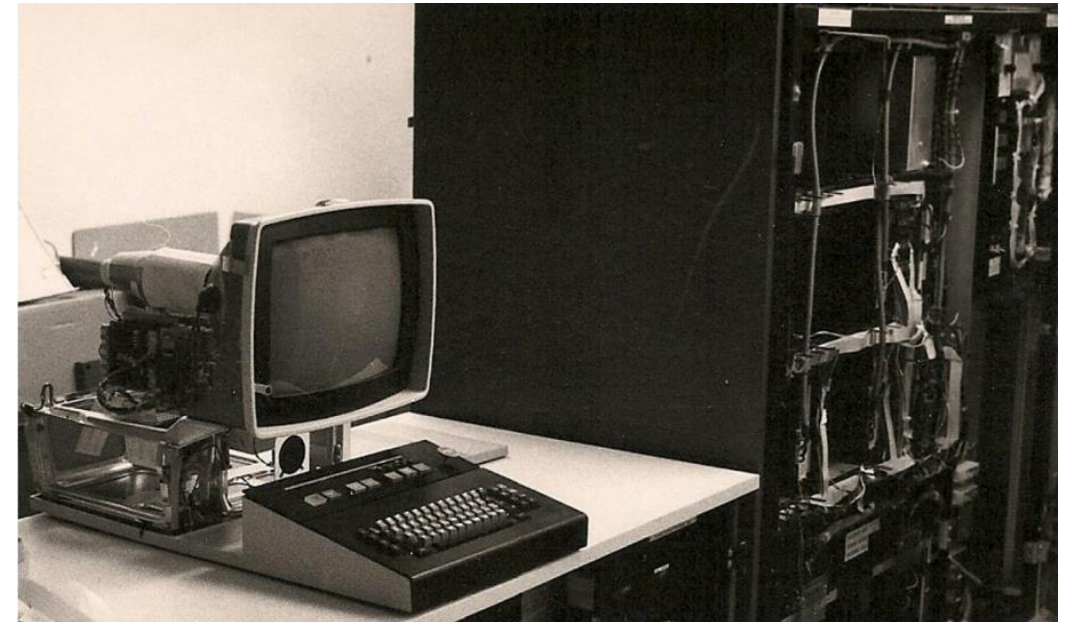
01

Intro – What the Hell is a Mainframe

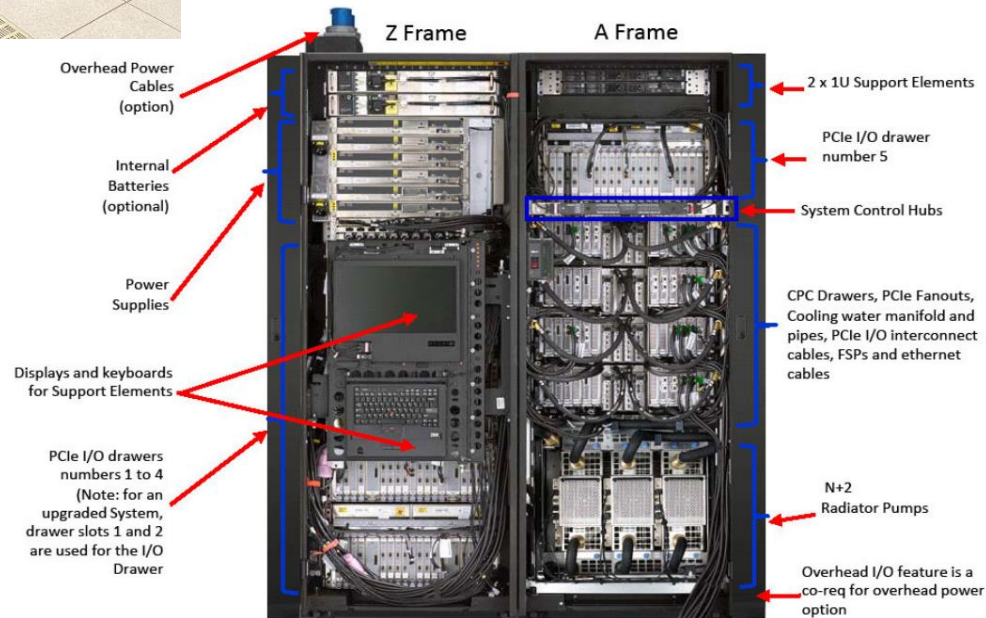
Mainframe – some history



- Start with Enterprise Computing in 1960/'70
 - Filled whole rooms
 - No visual device for input/output
 - First programming languages
 - Fortran (1958)
 - Cobol (1959)
- In the /'80s strong competition with mini-computer (today's midrange)
- '90 market consolidation and decline
- Several modernization initiatives
 - End of '90 a implementation of a UNIX-like system
 - 'til now running Docker containers



Mainframe – what does it look like?



Mainframe - hardware

- Categorized in machine types and models
- Every component at full capacity
 - Sub capacity is software managed
- Every component redundant
 - CPU, memory, network, power delivery ecc.
 - In case of failure automatic switch over
- Mainframes can be clustered (parallel sysplex)
- CISC vs RISC Architecture
 - microcode for dedicated tasks

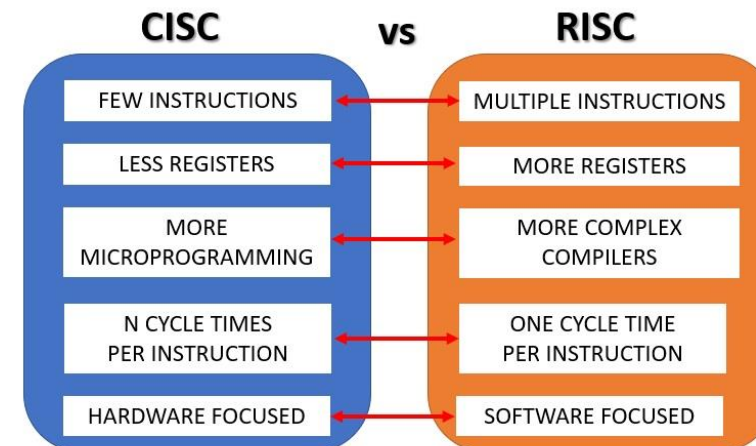


Mainframe Engine Types

- Central Processor (CP)
- Integrated Facility for Linux (IFL)
- Integrated Coupling Facility (ICF)
- System Assistance Processor (SAP)
- System z Application Assist Processor (zAAP)
- System z Integrated Information Processor (zIIP)

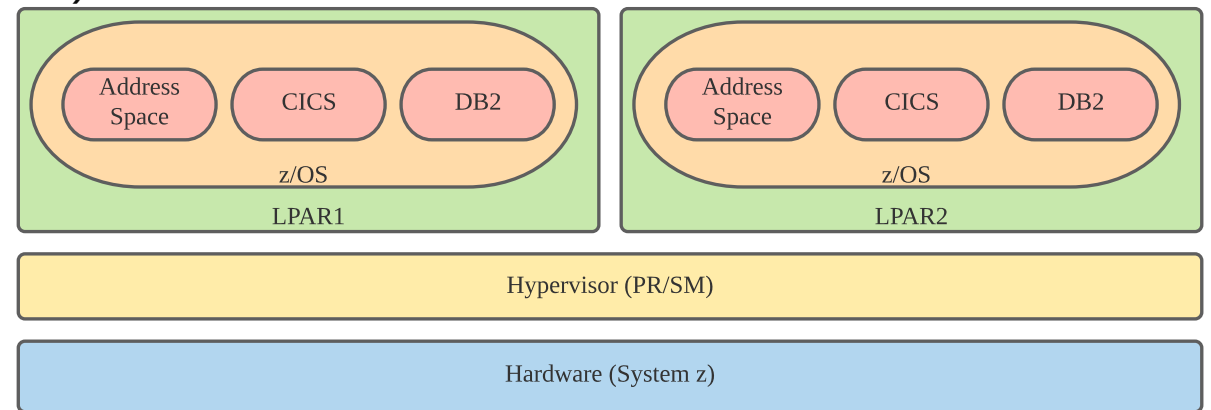
Many other specialty processors

- CPACF
- zEDC
- ...



Mainframe – System Software

- On hardware runs type 1 hypervisor => PR/SM
- Creates logical partitions (LPAR) => virtual machines
- OS runs in those machines => z/OS
- z/OS built from OS/390 (MVS) + Unix
- Many different processes => started tasks
 - CICS – (Customer Information Control System)
 - JES – (Job Entry Subsystem)
 - DB2 – (Database 2)



<https://mainframe2cloud.com/Virtualization-Journey-Mainframe-Cloud/>

<https://de.wikipedia.org/wiki/LPAR>

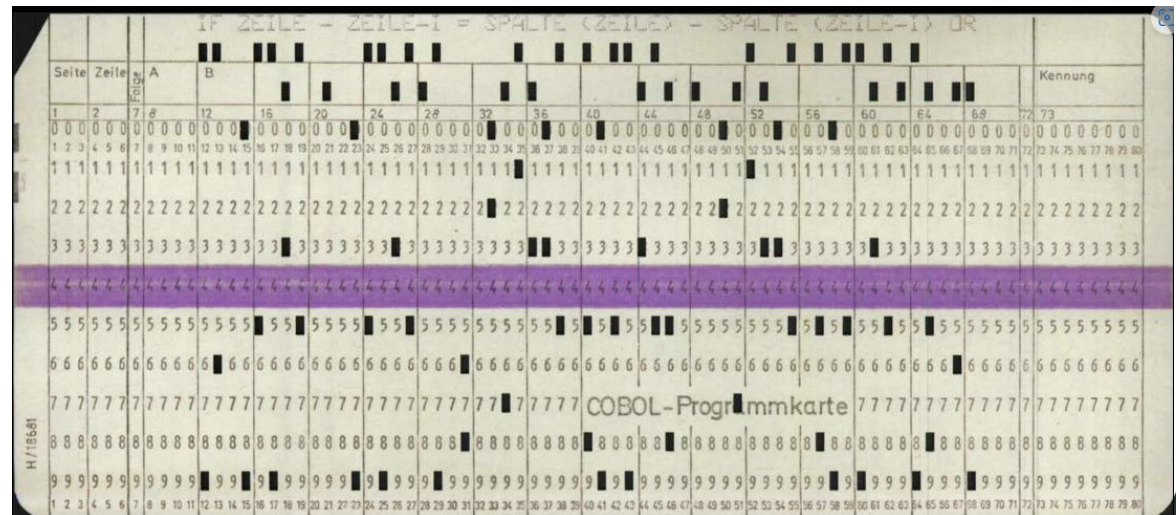
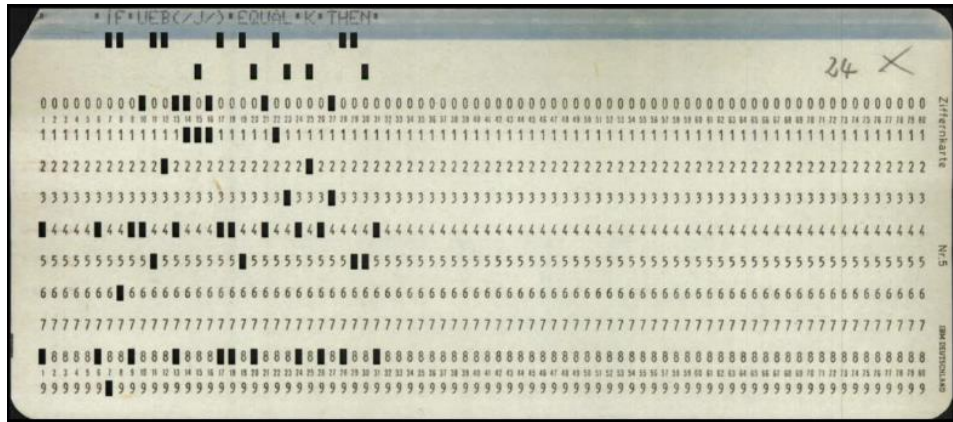
https://en.wikipedia.org/wiki/Logical_partition

02

Programming the Shark: The Basics

Programming – where does it come from?

- No visual input/output device
- No persistent storage device



Programming – that's so '90s

- TSO (Time sharing option)
 - System for interactive communication
 - Nothing else than a CLI
- ISPF (Interactive System Productivity Facility)
 - Is a grafical interface on top of TSO
- Without any licensing fees



```
3270 Terminal
Datei Bearbeiten Einstellungen Anzeigen Kommunikation Aktionen Fenster Hilfe

Menu Utilities Compilers Options Status Help
-----
ISPF Primary Option Menu

Option ==>
0 Settings      Terminal and user parameters
1 View          Display source data or listings
2 Edit          Create or change source data
3 Utilities      Perform utility functions
6 Command       Enter TSO or Workstation commands
8 DB2           Data Base 2
10 TLMS         Inquiry TLMS
Q QUICKREF      MVS/QUICK-REF
S SDSF          System Display and Search Facility
X EXIT          Terminate ISPF

Calendar
September 2024
Su Mo Tu We Th Fr Sa
1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

Time . . . . : 10:42
Day of year. : 263

Enter END command to terminate ISPF.

MA A 04/014
Verbindung zum fernen Server/Host tn3270.srv.raiffeisen.it aufgebaut über LU/Pool TCPRI513 und Anschluss 50023. \\vrspnt01-p\OG1P001 RICOH COLOR C3004 E
```

Programming – r u kidding me?

- Login was already done
 - Single sign on
- Starting point is ISPF main menu
- Steps performed
 - Choose library
 - Find member
 - Open program
 - Find position in code
 - Add line of code
 - Save member

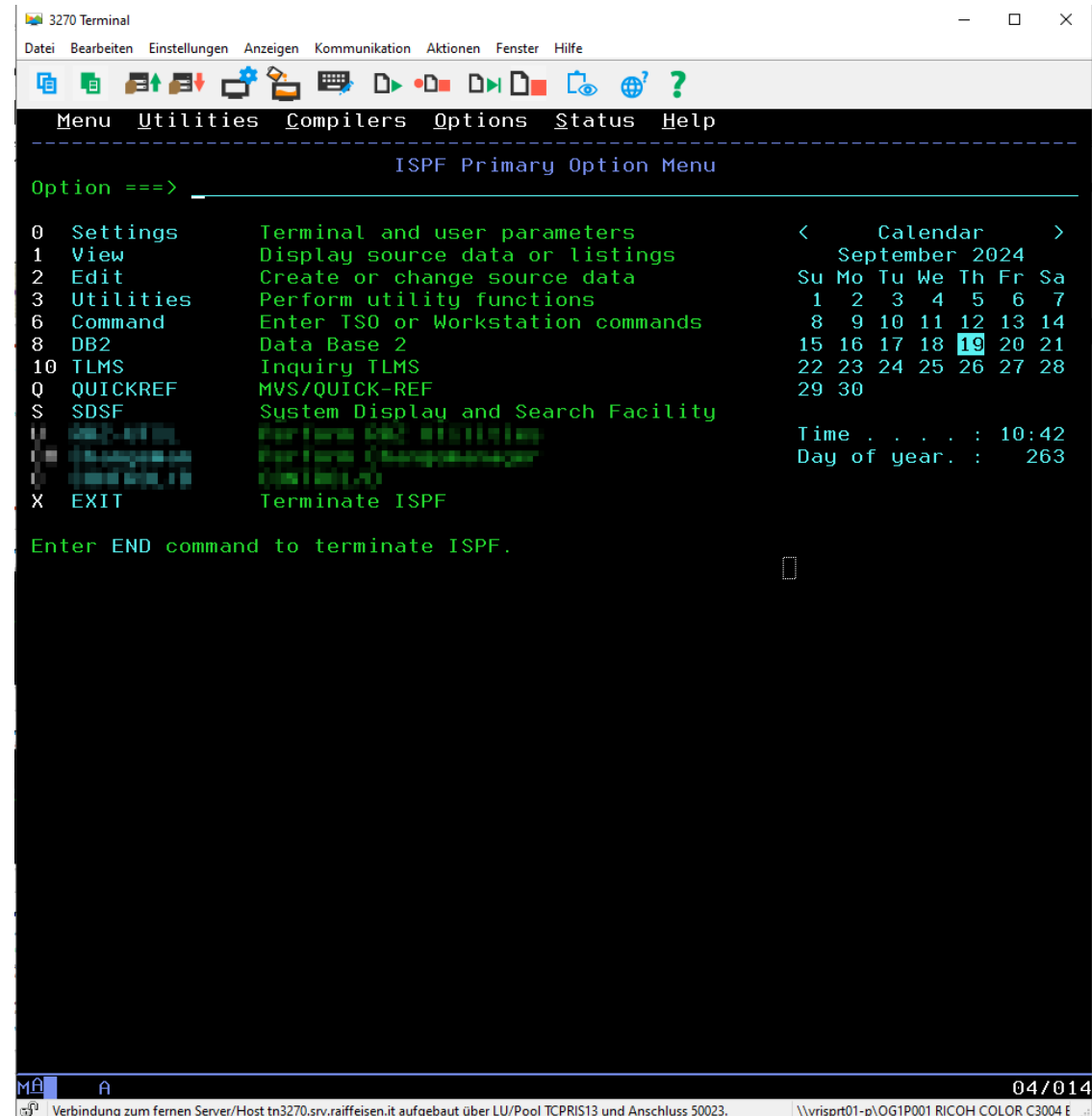
The screenshot shows a 3270 Terminal window titled "3270 Terminal". The window has a menu bar with "Datei", "Bearbeiten", "Einstellungen", "Anzeigen", "Kommunikation", "Aktionen", "Fenster", and "Hilfe". Below the menu bar is a toolbar with various icons. The main display area shows the "ISPF Primary Option Menu" with a list of options on the left and a calendar on the right. The options are:

Option	Description
0	Settings
1	View
2	Edit
3	Utilities
6	Command
8	DB2
10	TLMS
Q	QUICKREF
S	SDSF
X	EXIT

The calendar on the right shows the month of September 2024, with the 19th highlighted. The time is 10:42 and the day of the year is 263. At the bottom of the terminal, there is a status bar with the text "Verbindung zum fernen Server/Host tn3270.srv.raiffeisen.it aufgebaut über LU/Pool TCPRI513 und Anschluss 50023." and a small icon.

Programming – r u kidding me? Cont.

- Starting point ISPF main menu
- Steps performed
 - Choose library
 - Find member
 - Open JCL
 - Find position in job
 - Add line of code
 - Save member
 - Submit job



The screenshot shows a 3270 Terminal window titled "3270 Terminal" with a menu bar (Datei, Bearbeiten, Einstellungen, Anzeigen, Kommunikation, Aktionen, Fenster, Hilfe) and a toolbar. The main display area shows the "ISPF Primary Option Menu" with a list of options on the left and a calendar on the right. The options are:

Option	Description
0	Settings
1	View
2	Edit
3	Utilities
6	Command
8	DB2
10	TLMS
Q	QUICKREF
S	SDSF
X	EXIT

The descriptions for these options are: "Terminal and user parameters", "Display source data or listings", "Create or change source data", "Perform utility functions", "Enter TSO or Workstation commands", "Data Base 2", "Inquiry TLMS", "MVS/QUICK-REF", "System Display and Search Facility", and "Terminate ISPF".

At the bottom of the screen, it says "Enter END command to terminate ISPF." and "MA A".

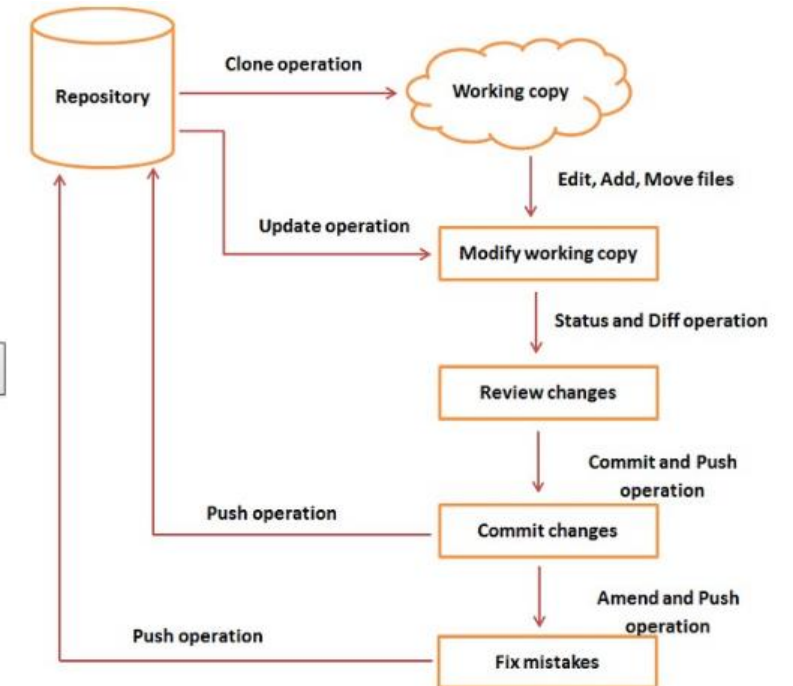
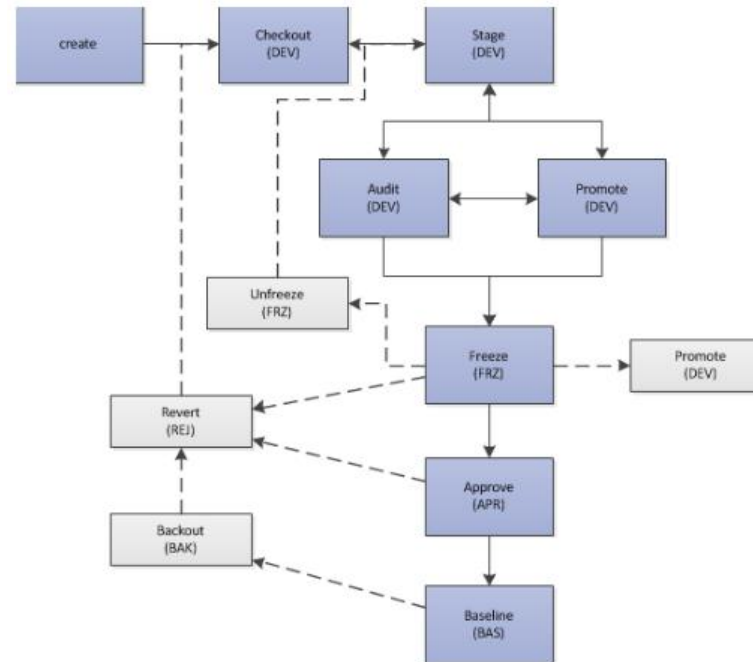
The status bar at the bottom shows the connection information: "Verbindung zum fernen Server/Host tn3270.srv.raiffeisen.it aufgebaut über LU/Pool TCPRIS13 und Anschluss 50023." and the date "04/014".

Programming – Java is my style!

- How was the name of the function? F***, where is my code completion?
- Yes, my variable is named foreignLoanFundingAmount!
- Really? Are there other string types like VARCHAR?
- I have to store a String in Db2. I use VARCHAR!
- Where is my Integer and Double data type?
- Of course I have to read the whole Db2 table!
- Why should I care about performance? My task is the business logic!
- My program is failing in production? What should I do with an offset?

Programming – Changeman what cont.?

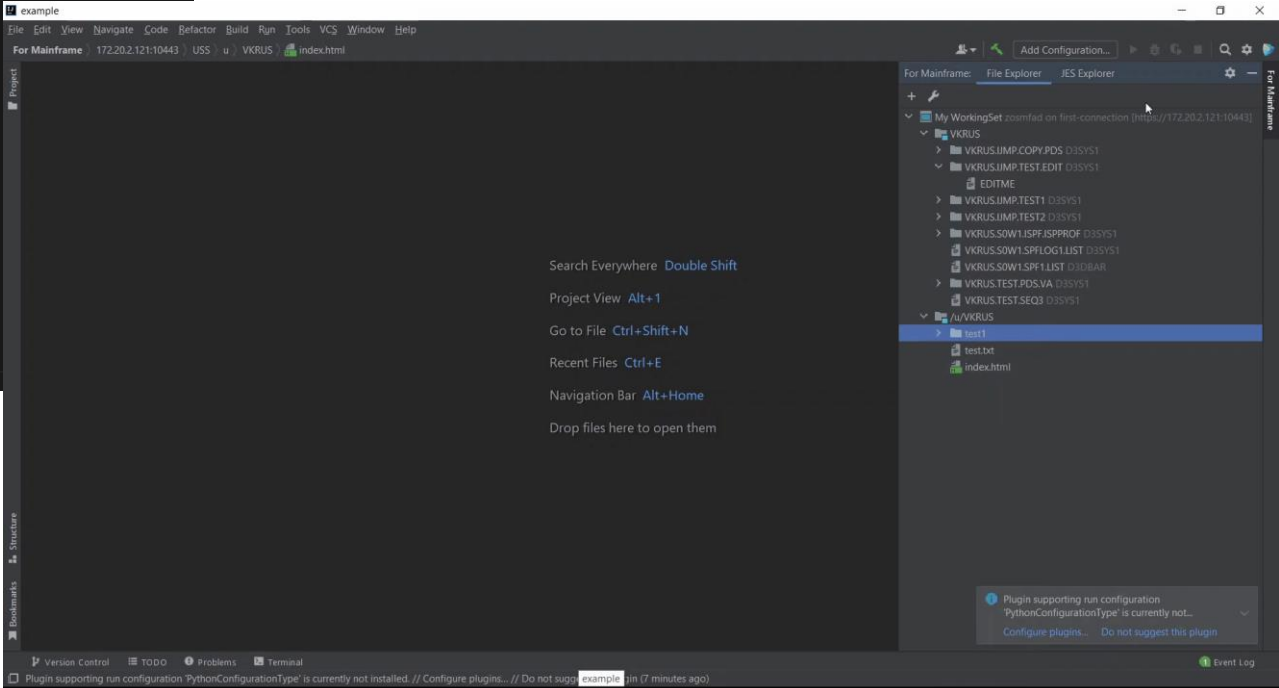
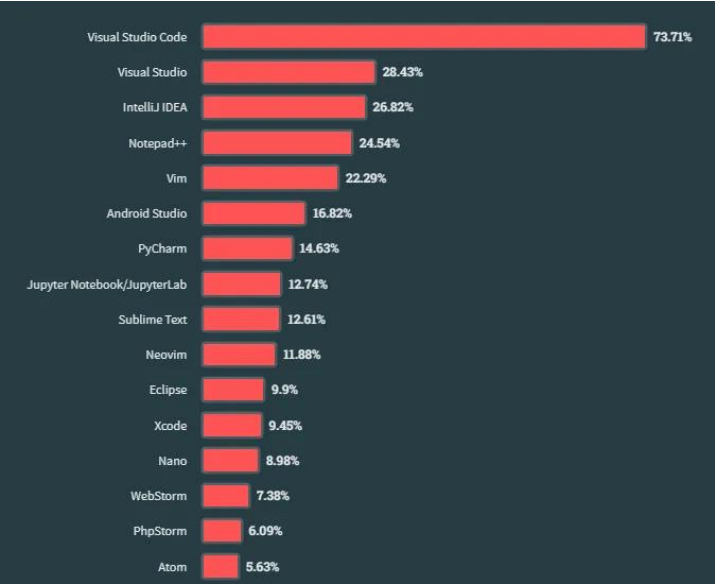
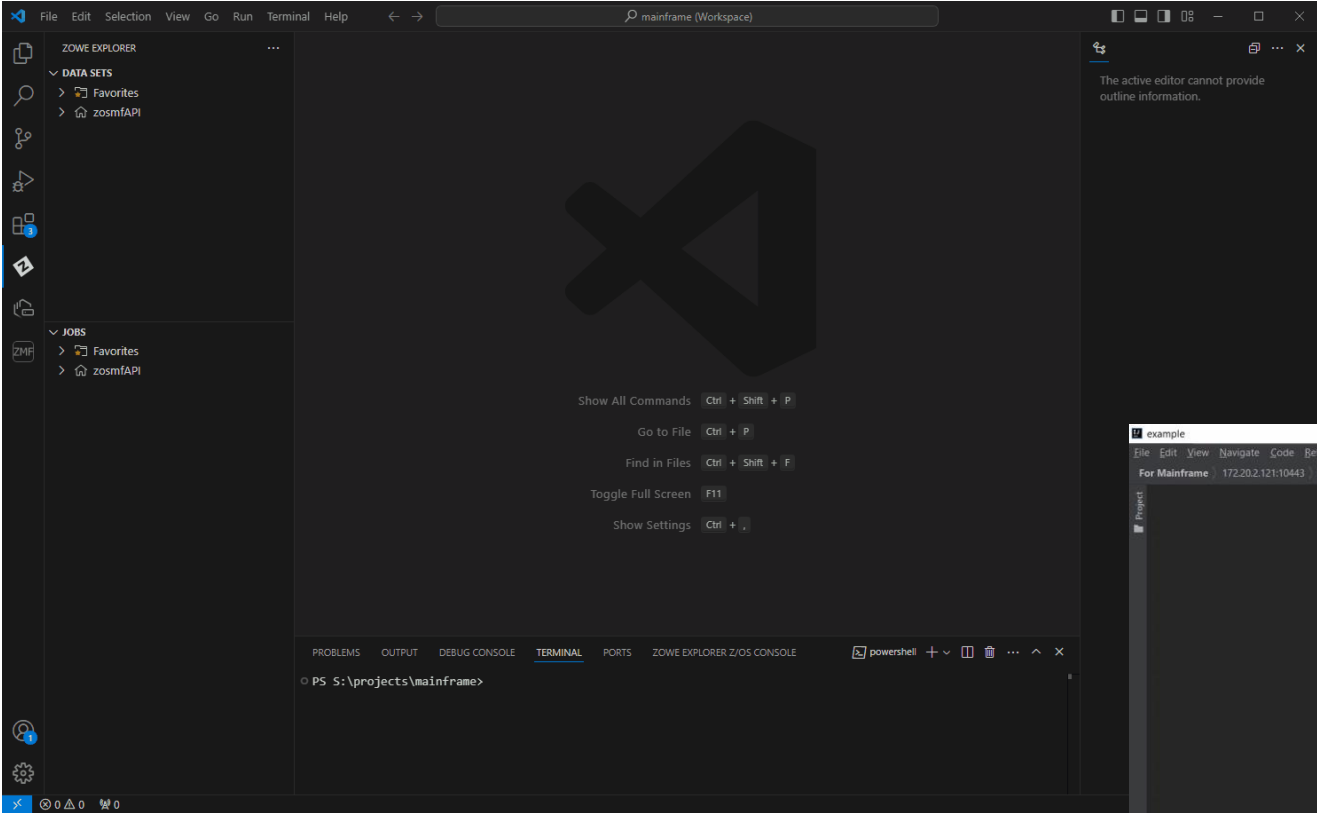
- Difference in ,checkout philosophy‘
 - CMN works on files
 - GIT works on whole branches
- Collaboration
 - CMN can connect package
 - Merge or Cherry pick



03

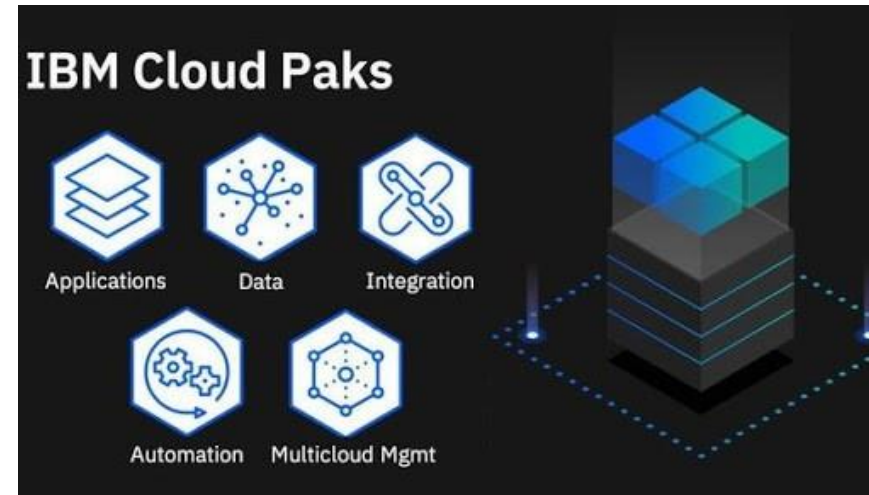
The Mainframe lives! Trends and Innovations

Trends - In programming: modern environments



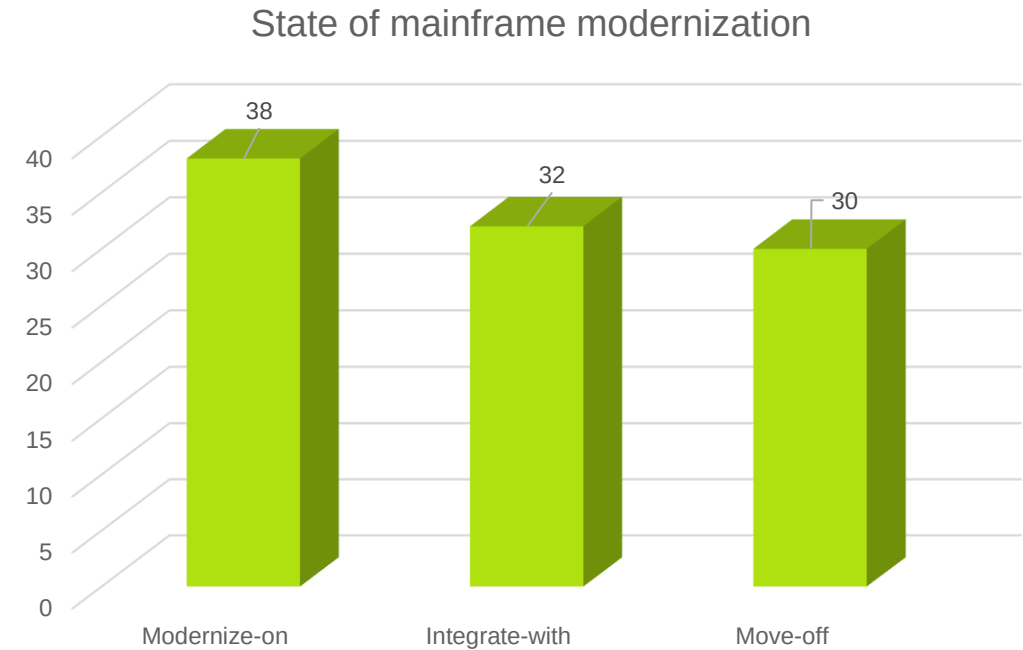
Trends - In core systems

- Hybrid cloud integrations
 - Precious Db2 data through data fabric
- Open up to the REST world
 - Online transactions as REST
 - Db2 Stored procedures as REST
- Consumable through Pipelines
 - Jenkins
 - GITLab CI
 - Bitbucket



Trends – In platform

- Embrace Open
 - NodeJS
 - Git
 - Ansible
 - ...
- Mainframe is part of the hybrid cloud
 - Many modernization initiatives
- Running container workload
 - zCX after aquisition of Redhat
- Enlarge observability
 - Use open standards
 - Integrate mainframe as ,normal server‘

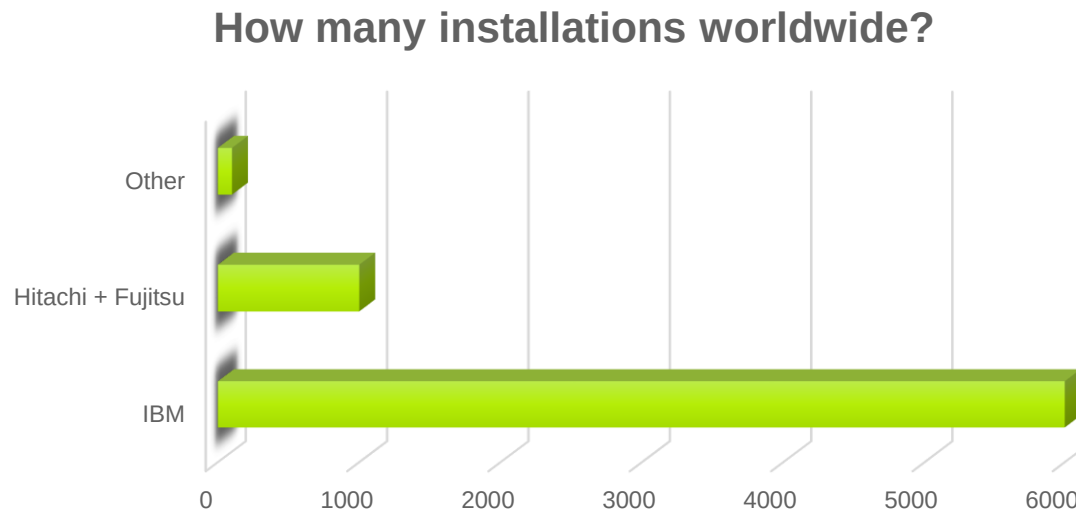


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Numbers and Insights – Data and Statistics

Numbers - Mainframe in numbers

- Mostly no exact numbers
- Data/statistics based on surveys



According to IBM

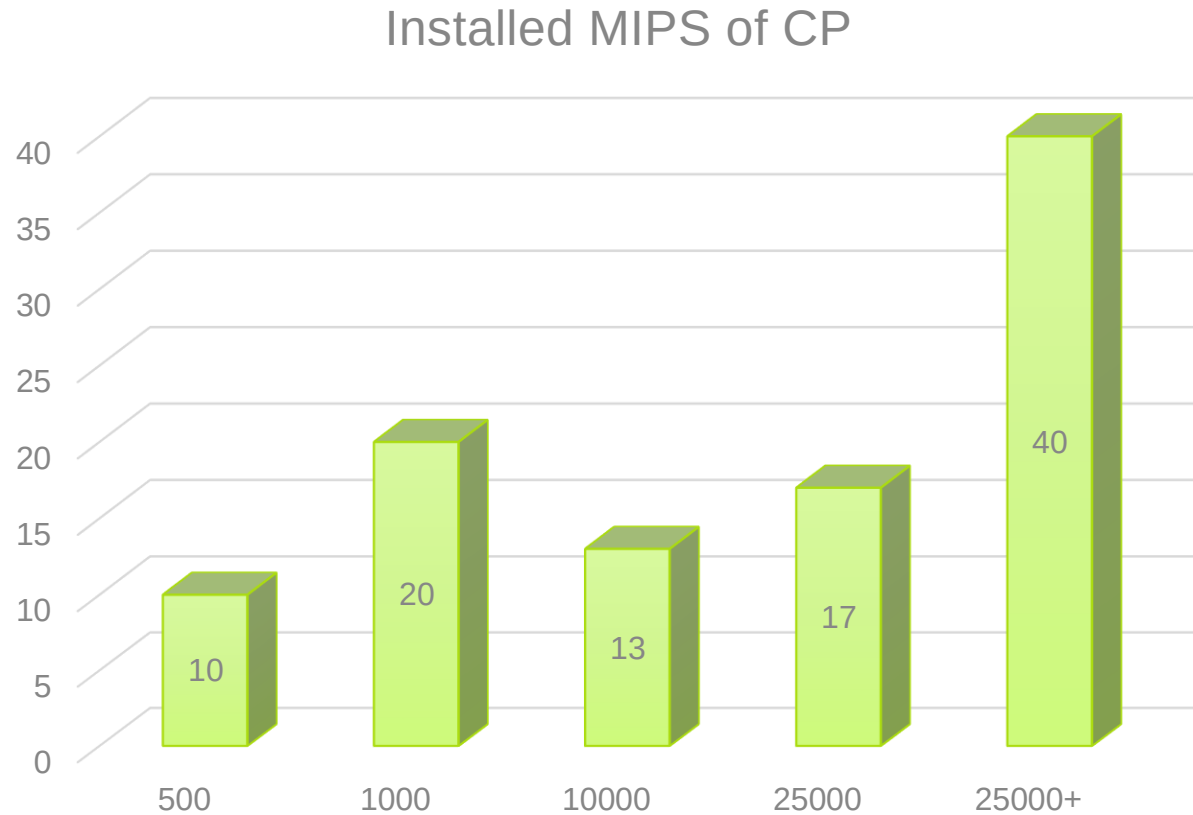
- They have 90% of market share on mainframe
- In 90% of the cases, a credit card payment goes through an IBM mainframe
- 92 of world's top 100 banks rely on mainframes
- 71% of Fortune 500 companies use IBM mainframes

Numbers - Who uses mainframes

- Financial services
 - Banks
 - Assurance companies
- Healthcare
- Government
 - Federal and State/local
- Online Retailer
- Transport and Logistics
- Energy
- Telecommunication



Numbers - How big are the installations?



Numbers – And in RIS?



IBM z15

2.400 MIPS

3 Processors

120 GB RAM

10.000 disks

4 environments

40 banks supported



4,5M Online req./day

Avg. resp. time 15ms



350 DB Tx/s

10K-15K DML/s

330 MB/s



30.000 Batch jobs

05

Q&A: Questions and Answers